NETWORKING & DNS QUICK NOTES

========== 1. NETWORK INTERFACES
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lo (Loopback Interface)

- IP: 127.0.0.1
- Used for communication within the same system.
- Computer talks to itself.

Example: ping 127.0.0.1

ens33

- Ethernet network interface.
- Used for communication with other systems and the Internet.

View Interfaces: ip addr ip link show

========== 2. IP COMMANDS
==========

ip addr

Shows IP addresses of interfaces.

ip route

Shows routing table.

Example: default via 192.168.1.1 dev ens33

Meaning: default gateway = 192.168.1.1

Add IP Address: sudo ip addr add 192.168.1.50/24 dev eth0

192.168.1.50 = IP Address /24 = Subnet Mask (255.255.255.0) eth0 = Interface

========== 3. CONNECTIVITY COMMANDS
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ping

Tests connectivity.

Example: ping google.com

traceroute

Shows path taken by packets.

Example: traceroute google.com

- ◦ ▪ in traceroute means:
- Router did not respond.
- Firewall may be blocking packets.
- Does NOT always mean network failure.

tracepath

Similar to traceroute. Also discovers PMTU (Path MTU).

Difference: traceroute -> Detailed tracepath -> Simpler + PMTU

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DNS = Domain Name System

Converts:

google.com ↓ 142.250.x.x

nslookup

Basic DNS lookup.

Example: nslookup google.com

dig

Most powerful DNS command.

Example: dig google.com

Useful: dig google.com MX dig google.com NS dig google.com TXT dig -x 8.8.8.8

host

Simple DNS lookup utility.

Example: host google.com

Interview Preference: dig > host > nslookup

========== 5. DNS RECORD TYPES
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A Record

Domain -> IPv4 Address

Example: google.com -> 142.250.x.x

AAAA Record

Domain -> IPv6 Address

MX Record

Mail Server Record

Example: gmail.com ↓ gmail-smtp-in.l.google.com

Used for email delivery.

NS Record

Name Server Record

Example: google.com ↓ ns1.google.com ns2.google.com

Tells which DNS servers manage the domain.

CNAME Record

Alias Record

Example: www.company.com ↓ company.com ↓ 192.168.1.10

Memory: A Record = Name -> IP CNAME Record = Name -> Name

PTR Record

Reverse DNS

Example: 8.8.8.8 ↓ dns.google

TXT Record

Stores text information.

Used for:

- SPF
- DKIM
- DMARC
- Domain Verification

========== 6. MX vs NS RECORD
==========

MX Record

Handles Email Delivery.

Example: user@gmail.com ↓ gmail-smtp-in.l.google.com

NS Record

Handles DNS Queries.

Example: google.com ↓ ns1.google.com

Memory:

MX = Mail Server NS = DNS Server

========== 7. EMAIL AUTHENTICATION
==========

Purpose: Prevent Email Spoofing

Technologies:

1. SPF
2. DKIM
3. DMARC

========== 8. SPF ==========

Full Form: Sender Policy Framework

Purpose: Checks whether sending mail server is authorized.

Question Answered: "Is this mail server allowed to send emails for this domain?"

SPF checks: ✔ Sender Mail Server IP

SPF does NOT check: ✘ Human identity ✘ Email content

Memory: SPF = Who sent the email?

========== 9. DKIM ==========

Full Form: DomainKeys Identified Mail

Purpose: Verifies email integrity using digital signatures.

Checks:

- Was email modified?
- Was it signed by domain owner?

Uses: Private Key Public Key

Memory: DKIM = Was email changed?

========== 10. DMARC ==========

Full Form: Domain-based Message Authentication, Reporting and Conformance

Purpose: Defines action when SPF/DKIM fail.

Policies:

p=none p=quarantine p=reject

Memory: DMARC = What action should be taken?

========== 11. TCP CONNECTION STATES ==========

TIME_WAIT

- You closed connection first.
- TCP waiting for safety.
- Usually normal.

CLOSE_WAIT

- Remote side closed first.
- Application has not closed socket.

Memory:

TIME_WAIT -> Usually Normal

CLOSE_WAIT -> Application Problem

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12. SS COMMAND
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Modern replacement for netstat.

Most Important:

ss -tulpn

Options:

t = TCP u = UDP l = Listening p = Process n = Numeric a = All x = UNIX Sockets s = Summary e = Extended o = Timers 4 = IPv4 6 = IPv6

Useful Commands:

ss -tuln Shows listening ports

ss -tulpn Shows listening ports + process names

ss -tan Shows all TCP connections

ss -x Shows UNIX sockets

ss -s Shows summary statistics

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13. UNIX SOCKETS
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Used for: Process-to-Process communication on same Linux machine.

Examples:

/run/dbus/system_bus_socket /run/user/1000/wayland-0

Difference:

TCP Socket

Communication between machines

UNIX Socket

Communication within same machine

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14. INTERVIEW ONE-LINERS
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A Record: Domain -> IP Address

CNAME: Domain -> Another Domain

MX: Domain -> Mail Server

NS: Domain -> DNS Server

PTR: IP Address -> Domain

SPF: Verifies sender mail server

DKIM: Verifies email signature

DMARC: Defines action if SPF/DKIM fail

TIME_WAIT: Normal TCP closing state

CLOSE_WAIT: Application not closing socket

ss -tulpn: Shows open ports and processes

dig: Most powerful DNS troubleshooting tool